

Omar Muhammad

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EDUCATION

B.Tech. in Mathematics and Computing | 2023 – 2027 | Indian Institute of Science, Bengaluru | CGPA: 9.5/10.0

Grade 12 (CBSE) | 2022 – 2023 | Saraswathi Vidyanikethan, Ernakulam | 98.4% (School Topper)

Grade 10 (CBSE) | 2020 – 2021 | Rajagiri Public School, Ernakulam | 98%

PUBLICATIONS AND PREPRINTS

Verification Modulo Tested Library Contracts

Abhishek Uppar, **Omar Muhammad**, Sumanth Prabhu S, Deepak D'Souza, P. Madhusudan, Adithya Murali.

Accepted to ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI 2026) and appeared in PACMPL. [\[Paper\]](#) [\[Code\]](#)

Have a thing? Reasoning around recursion with dynamic typing in grounded arithmetic

Bryan Ford, Elliot Bobrow, Stefan Milenković, **Omar Muhammad**, Dimitrios Alexopoulos, Yusuf Demir, Ananthajit Srikant

Under review at the ACM SIGPLAN Conference on Principles of Programming Languages. [\[Paper\]](#)

Mask2Cause: Temporal Causal Discovery Beyond Causality in Mean

Omar Muhammad, Pasupuleti Dhruv Shivkant, Deepak N. Subramani

Under review at the International Conference on Learning Representations (ICLR 2027). [\[Paper\]](#) [\[Code\]](#)

Phase Transition Thresholds when Finite Automata Play Supergames under Finite Costs for Complexity

Omar Muhammad, Amrit Mahendra Joseph

Poster submission under review at Workshop on Research Highlights in Programming Languages (RHPL 2026)

RESEARCH EXPERIENCE

Summer Research Fellow, Grounded Deduction | EPFL, Switzerland (Onsite for May – July 2026; Remote after) | May 2026 – Present

Adviser: Prof. Bryan Ford, EPFL

- Developing Grounded Deduction, a formal reasoning framework that admits arbitrary recursive definitions without prior termination proofs, using explicit definedness conditions to reason about potentially nonterminating computations. [\[Slides\]](#)
- Building on an existing Isabelle/Pure implementation of Grounded Deduction, encoded Basic Grounded Arithmetic's syntax, semantics, and proof checker and proved its consistency within GD, using step-indexed satisfaction.
- Developing GD2, a usability-focused implementation of Grounded Deduction in Isabelle/Pure, with unified terms and formulas, first-class functions, and a definable fixed-point operator supporting recursive and mutually recursive definitions without new axioms or termination proofs at definition time.
- Adapting and extending proof automation and libraries, with comparative case studies in Isabelle/HOL and Lean 4; developing a hammer that translates GD2 goals to first-order problems and reconstructs external proofs in Isabelle.
- Extended GD's mechanized metatheory by adding proof-checker rules and proving primitive recursiveness and truth preservation; proved that Propositional Grounded Arithmetic (PGA) contains its own truth predicate.

Student Researcher, Verification Modulo Tested Library Contracts | IISc | May 2025 – Apr 2026

Advisers: Prof. D. D'Souza (IISc), Prof. P. Madhusudan (UIUC), Prof. A. Murali (UW-Madison)

- Co-developed VMTLC, a framework that verifies client programs assuming synthesized library contracts, while testing those contracts against the library implementations.
- Co-designed the Learner component within a Teacher-Learner CEGIS loop, developing two distinct LLM-integrated architectures (a pure LLM learner and an LLM-seeded ICE learner) to leverage semantic intuition, synthesizing valid contracts where traditional enumerative baselines could not.
- Independently designed and implemented the LLM pipeline for synthesizing contracts and invariants from logical constraints, including prompt templates, counterexample-driven repair, malformed-output handling, and integration with formal verification and testing; implemented LLM-generated attributes for the hybrid HornICE learner.
- Designed and implemented two mutation templates producing 30 incorrect programs that escaped testing; analyzed when modular and contextual VMTLC rejected or accepted them.

Student Researcher, Causal Discovery in Time Series | IISc | Aug 2025 – Present

Adviser: Prof. Deepak N. Subramani, Computational and Data Sciences, IISc

- Led the research and designed and implemented Mask2Cause, a Transformer framework for causal discovery through conditional mean and variance prediction using adjacency-constrained attention over inverted variable embeddings.
- Designed a variance-aware prediction objective and the novel Mixed Physics benchmark to study dependencies expressed through conditional mean and variance; developed an information-theoretic analysis motivating the objective.
- Designed controlled experiments investigating information leakage, hidden confounding, changing causal graphs, and separation of mean and variance parents; used the findings to refine architectural claims and characterize limitations.

Student Researcher, Finite Automata in Repeated Games | Jan 2026 – Present | IISc

- Leading a collaborative investigation of repeated Prisoner's Dilemma strategies implemented as finite automata, studying how per-state memory costs affect cooperation and equilibrium.
- Derived a tight cost threshold for sustaining eventual mutual cooperation in Nash equilibrium, and bounds on when non-defection outcomes disappear under Nash and semi-perfect equilibrium.

Summer Research Fellow, Formalization of Group Theoretic Algorithms | Krea University, India (Onsite) | May 2025 – July 2025

Advisers: Prof. Rishi Vyas (KREA), Prof. T.V.H Prathamesh (KREA)

- Formalized proofs for decision problems in geometric group theory (including the word, conjugacy, and subgroup membership problems) using the Lean 4 proof assistant.
- Key contributions include formalizing two definitions of the Dehn function and proving their equivalence and verifying the algorithm for the conjugacy problem in free groups.
- The project forms part of a broader research effort on formalizing algorithms in geometric group theory, including the computability of Dehn functions for hyperbolic groups, for which I helped formalize the foundations.

Research Intern, Enhancing Formal Verification with LLMs | University of Virginia, USA (Remote) | Aug 2025 – Oct 2025

Adviser: Prof. Wenxi Wang, The University of Virginia

- Explored techniques for decomposing Z3 proof objects for Verus-generated verification conditions into dependency-preserving proof subtrees, aiming to make solver reasoning easier for LLMs to interpret.
- Assisted with evaluation of LLM-generated verification proofs and literature review; contributions acknowledged in the resulting ICML 2026 paper introducing VCoT-Lift and VCoT-Bench.

TEACHING

Teaching Assistant, Program Analysis and Verification | IISc | Aug 2026 – Present

Instructors: Prof. Deepak D'Souza, Prof. K. V. Raghavan

- Designed the course project and mentored student teams through it. Graded assignments, conducted vivas, and held regular office hours for doubt clearing.

PROJECTS

Independent Study, Logic and Automated Theorem Proving | IISc | Aug 2026 – Present

Adviser: Prof. Deepak D'Souza, Computer Science and Automation, IISc

- Studying completeness of first-order logic, Gödel's incompleteness theorems, Löb's theorem and Tarski's undefinability theorem. [\[Notes: Completeness of First-Order Logic\]](#)
- Exploring structural proof theory and automated proof search techniques.

Reading Project, Threshold for Random k -SAT | IISc | Jan 2026 – Apr 2026

Adviser: Prof. Riddhipratim Basu, ICTS-TIFR

- Explored the satisfiability threshold for random k -SAT as part of the "Techniques in Discrete Probability" course.
- Analyzed the foundational work by Dimitris Achlioptas and Yuval Peres, on the application of weighted moment methods to bound the threshold. [\[Report\]](#) [\[Presentation\]](#)

Reading Project, PCP Theorem | IISc | Aug 2025 – Nov 2025

Adviser: Prof. Anand Louis, Computer Science and Automation, IISc

- Conducted an in-depth study of the PCP theorem, analyzing its formulations in terms of both probabilistically checkable verifiers and the hardness of approximation.
- Analyzed Irit Dinur's proof of the theorem using gap amplification. [\[Report\]](#)

Formalization Project, Rubik's Cube Solvability Conditions | Feb 2025 – May 2025

- Modeled the $n \times n \times n$ Rubik's Cube as a group, formalized proof of validity of solvability conditions of the $4 \times 4 \times 4$ Rubik's Cube and formulated the solvability conditions for the $n \times n \times n$ cube in Lean 4. [\[GitHub Repo\]](#)

Reading Project, Undecidability of First-Order Logic | Feb 2025 – April 2025

- Studied the foundational proof by Alonzo Church, and its implications for computability and formal verification systems.

Research Project, Image Denoising Algorithms | March 2025 – April 2025

- Implemented and evaluated various frameworks for image restoration without clean data.
- Designed and conducted experiments to analyze the hyperparameter sensitivity of Zero-Shot Noise2Noise implementation. [\[Report\]](#) [\[Presentation\]](#)

Reading project, Differential Topology and Knot Theory | IISc | May 2024 – July 2024

Prof. Subhojoy Gupta, Mathematics, IISc

- Explored topics in Differential Topology and Knot Theory.

Developer DataBased (CS club at IISc) | Oct 2023 – Dec 2023

- Utilized Generative AI to develop components for an automated Refute Problem Generation system.

SOFTWARE

Intern | National Payments Corporation of India (NPCI) | Dec 2024 – June 2025

Adviser: Prof. Viraj Kumar, Divecha Centre for Climate Change, IISc

- Developed an internal tool to enable NPCI employees to query databases and generate data visualizations via natural language prompts.
- Implemented a pipeline that translates natural language queries to SQL queries, fetches data, determines correct visualization types, and renders visual output.

Intern | Applied Research Capability Network (ARC-Net), IISc Campus | July 2024 – Jan 2025

- Fine-tuned LLMs with Tool Integrated reasoning for solving math and physics problems.
- Used proof assistants and symbolic solvers to verify the chain of thought of the models.

ACHIEVEMENTS AND RECOGNITIONS

- Selected for MITACS Globalink Research Internship 2026
- JEE Advanced Examination: All India Rank 646
- JEE Mains Examination: 99.95 percentile, All India Rank 654
- KVPY (Kishore Vaigyanik Protsahan Yojana) Scholar
- NTSE (National Talent Search Examination) Scholar
- NSEP (National Standard Examination in Physics) - National top 1%

TECHNICAL SKILLS

- **Verification & Logic:** Isabelle, Lean 4, Z3, Alloy, Spin, Rodin, VCC, JPF, Verus, SootUp
- **Machine Learning:** PyTorch, Scikit-learn, Transformers, LLM Fine-Tuning
- **Languages & Tools:** Python, C, SML, SQL

SELECTED COURSEWORK

- **Theoretical Computer Science & Formal Methods:** Proofs and Programs; Automata and Computability; Program Analysis and Verification; Formal Methods in Software Engineering; Theorist's Toolkit; Game Theory
- **Mathematics & Machine Learning:** Probability and Statistics; Techniques in Discrete Probability; Applied Data Science and Artificial Intelligence; Introduction to Algebraic Structures

EXTRACURRICULAR ACTIVITIES

- Mentored underprivileged students in STEM concepts through the Notebook Drive Mentorship Outreach Program, IISc.
- Active Member of DataBased (Computer Science Club at IISc).
- Designed competitive programming problems for the RevCoding Contest as a Volunteer for Pravega 2025 (IISc UG Fest).
- Presented on 'Online Decision-Making Algorithms' at Algorithm Festival 2024, IISc.

LANGUAGES

- **English:** Fluent; IELTS Academic Overall: 8.0/9.0
- **Malayalam:** Native
- **Hindi:** Intermediate